

DAY-5

Tasks

1.

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	What to do	What you are checking
1. Check the wall socket	Connect another working device to the socket	Whether the power source is working
2. Inspect the charger	Check the adapter, cable, and connector for cuts, burns, or loose connections	Whether the charger is physically damaged
3. Check the charging indicator	Plug in the charger and look for the charging/power LED	Whether the laptop is receiving power
4. Perform a power reset	Disconnect the charger, remove the battery if removable, hold the power button for 15–30 seconds, then reconnect power	Whether a temporary power-state problem is preventing startup
5. Test the charger/battery	Test the charger's output with a multimeter or try a known-good compatible charger	Whether the charger or battery is faulty

2.

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

Troubleshooting flow

1. Check whether the monitor is powered on.
 - Make sure the power LED is on.
 - Check the monitor's power cable.
2. Check the display cable.
 - Make sure HDMI, DisplayPort, or VGA is firmly connected at both ends.
 - Look for damaged cables.
3. Check the monitor input/source.

- Open the monitor's input menu.
- Select the correct source, such as HDMI or DisplayPort.
- 4. Check where the cable is connected.
 - If the PC has a dedicated GPU, connect the monitor to the GPU's display port rather than the motherboard's port.
- 5. Restart the PC.
 - A temporary startup or display problem may be causing the blank screen.
- 6. Test another cable or monitor.
 - This helps determine whether the problem is the monitor/cable or the PC.
- 7. Check the GPU.
 - Make sure the graphics card is properly seated.
 - Check that its PCIe power cables are connected.
- 8. Check display settings.
 - If Windows starts but the display is incorrect, use Windows + P and select the appropriate display mode.
 - Check the resolution and refresh rate if necessary.
- 9. Confirm the fix.
 - The monitor should display the normal Windows desktop without flickering or loss of signal.

3.

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Step 1: Check the mouse itself

- Turn the mouse off and on.
- Check whether the batteries are charged.
- Replace the batteries if necessary.
- Check whether the mouse's sensor light is working, if applicable.

This determines whether the mouse itself has power.

Step 2: Check the wireless receiver

- Remove the USB receiver.

- Reconnect it firmly.
- Check whether it is physically damaged.

If possible, test the receiver on another USB port.

Step 3: Test another USB port

Move the mouse receiver to a different USB port.

- If it works → the original USB port may have a problem.
- If it doesn't work → continue troubleshooting.

Step 4: Test the mouse on another computer

Connect the wireless receiver to another computer.

- Works on another PC → the original system or its settings may be the problem.
- Doesn't work → the mouse/receiver is likely faulty.

Step 5: Check system settings

In Windows, check:

- Settings → Bluetooth & devices → Devices
- Device Manager → Mice and other pointing devices
- Check whether the mouse is detected or has a warning symbol.

You can also restart the computer and reinstall/update the mouse driver if necessary.

4.

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.
Hint: Think about fans, dust, and thermal paste.

Cause 1: Dust blocking airflow

Possible symptom: Fans are running, but the PC becomes very hot during gaming.

How to check:

1. Turn off and unplug the PC.
2. Open the case.
3. Inspect the CPU cooler, GPU, case fans, and air vents.
4. Look for dust buildup.
5. Clean the dust using appropriate compressed air.

Blocked airflow prevents hot air from leaving the case.

Cause 2: Faulty or non-working fans

Possible symptom: A fan isn't spinning or is making unusual noises.

How to check:

1. Start the PC and visually inspect the fans.
2. Check the CPU cooler fan.
3. Check the GPU fans.
4. Check the case fans.
5. Make sure their power connectors are properly connected.

A failed CPU fan can cause the CPU to overheat very quickly.

Cause 3: Old or poorly applied thermal paste

Possible symptom: CPU temperature becomes extremely high even though the fan is working and the PC is clean.

How to check:

1. Turn off and unplug the PC.
2. Remove the CPU cooler carefully.
3. Inspect the thermal paste between the CPU and cooler.
4. If the paste is dried out or insufficient, clean the old paste.
5. Apply fresh thermal paste correctly.
6. Reinstall the cooler firmly.

Final check

After fixing the suspected problem, run the gaming system again and check whether it remains stable without overheating or automatically shutting down.